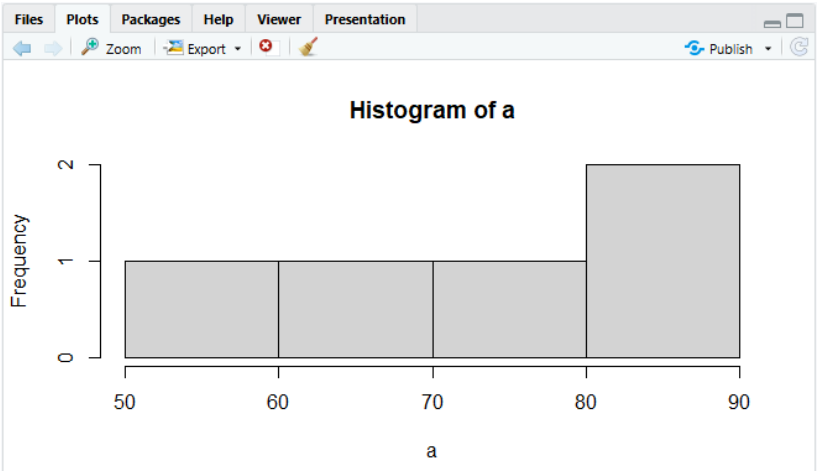
R-PROGRAMMING

16. HISTOGRAM

```
> a <- c(55, 67, 89, 80, 90)
>
> hist(a)
>
```



17. CORRELATION ANALYSIS

```
> library(readxl)
>
> diabetest1 <- read_excel("C:/Users/ACER/Downloads/NARA.xlsx")
>
> tab <- table(diabetest1$Age, diabetest1$Insulin)
> print(tab)

    80 85 88 90 95 98 100 105 110 115
21   1  0  0  0  0  0   0   0   0   0
23   0  1  0  0  0  0   0   0   0   0
24   0  0  0  0  1  0   0   0   0   0
25   0  0  1  0  0  0   0   0   0   0
27   0  0  0  1  0  0   0   0   0   0
28   0  0  0  0  0  0   1   0   0   0
29   0  0  0  0  0  0   0   0   1   0
30   0  0  0  0  0  1   0   0   0   0
31   0  0  0  0  0  0   0   1   0   0
33   0  0  0  0  0  0   0   0   0   1
35   0  0  0  0  0  0   0   0   0   0
36   0  0  0  0  0  0   0   0   0   0
38   0  0  0  0  0  0   0   0   0   0
39   0  0  0  0  0  0   0   0   0   0
41   0  0  0  0  0  0   0   0   0   0
42   0  0  0  0  0  0   0   0   0   0
44   0  0  0  0  0  0   0   0   0   0
45   0  0  0  0  0  0   0   0   0   0
47   0  0  0  0  0  0   0   0   0   0
50   0  0  0  0  0  0   0   0   0   0

    120 125 130 135 140 145 155 160
21    0   0   0   0   0   0   0   0
23    0   0   0   0   0   0   0   0
24    0   0   0   0   0   0   0   0
25    0   0   0   0   0   0   0   0
27    0   0   0   0   0   0   0   0
28    0   0   0   0   0   0   0   0
29    0   0   0   0   0   0   0   0
30    0   0   0   0   0   0   0   0
31    0   0   0   0   0   0   0   0
33    0   0   0   0   0   0   0   0
```

```
35    0    0    1    0    0    0    0    0
36    0    1    0    0    0    0    0    0
38    1    0    0    0    0    0    0    0
39    0    0    0    1    0    0    0    0
41    0    0    0    0    1    0    0    0
42    0    0    0    0    0    1    0    0
44    0    0    0    0    0    0    1    0
45    0    0    0    0    0    0    0    1
47    0    0    0    0    0    0    0    0
50    0    0    0    0    0    0    0    0

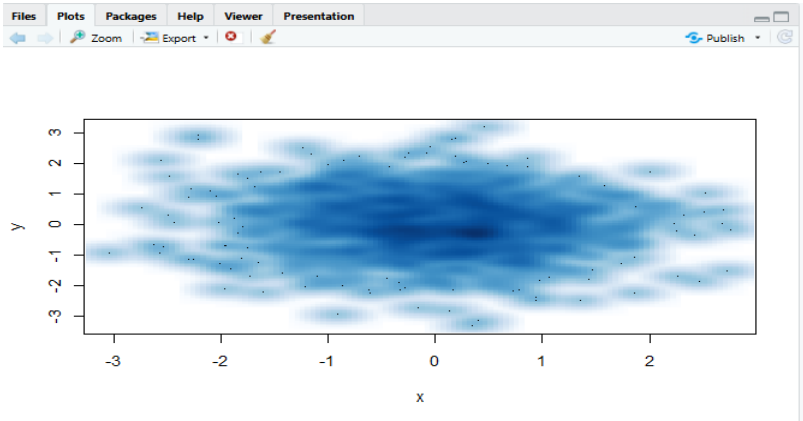
    165 175
21    0   0
23    0   0
24    0   0
25    0   0
27    0   0
28    0   0
29    0   0
30    0   0
31    0   0
33    0   0
35    0   0
36    0   0
38    0   0
39    0   0
41    0   0
42    0   0
44    0   0
45    0   0
47    1   0
50    0   1
>
> chisq.test(tab)

        Pearson's Chi-squared test

data:  tab
X-squared = 380, df = 361, p-value
= 0.2358
```

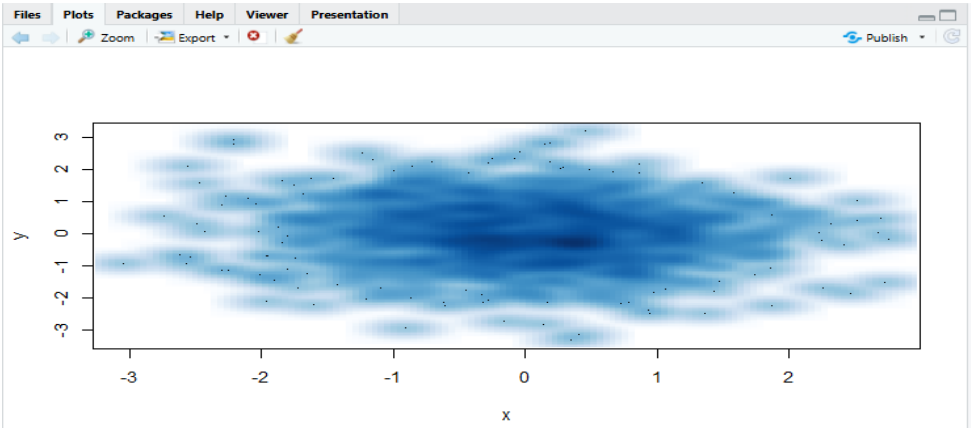
18. SCATTER PLOT

```
>
> set.seed(9)
> x<- rnorm(1000)
> y<- rnorm(1000)
> smoothScatter(y - x)
> smoothScatter(x,y)
>
```



19. LINEAR REGREESION

```
          2
> Relation <- lm(diabetes$BloodPressure ~ diabetes$Age)
> Png <- png("linear regression.png")
>
> plot(diabetes$Age, diabetes$BloodPressure, col="green",
+      main="Linear Regression Analysis",
+      abline=lm(diabetes$BloodPressure ~ diabetes$Age),
+      xlab="BloodPressure", ylab="Age")
```



20. MULTIPLE REGRESSION

```
> Input <- diabetes[, c("Age", "BloodPressure", "Glucose")]
> Model <- lm(Age ~ BloodPressure + Glucose, data = Input)
> print(Model)

Call:
lm(formula = Age ~ BloodPressure + Glucose, data = Input)

Coefficients:
  (Intercept)  BloodPressure         Glucose
      -0.4523        -0.2206          0.4028

> A<- coef(Model)
> print(A)
  (Intercept) BloodPressure       Glucose
   -0.4523418    -0.2205621     0.4028252
> xBloodPressure <- coef(Model)[2]
> yGlucose <- coef(Model)[3]
>
> print(xBloodPressure)
BloodPressure
   -0.2205621
> print(yGlucose)
  Glucose
0.4028252
> y = A+xBloodPressure + yGlucose
> print(y)
  (Intercept) BloodPressure       Glucose
  -0.27007869   -0.03829893    0.58508836
>
```